



# YAHYA AHMET OGUTCU

## ELECTRICAL AND ELECTRONICS ENGINEER

📍 Istanbul, Turkey

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I graduated from Bursa Technical University with a degree in Electrical and Electronics Engineering, achieving a GPA of 3.09/4.00. I have a strong interest in power electronics, electromagnetism, and communications, and I attach great importance to enhancing my theoretical knowledge through real-world projects and practical applications. In addition to my internship experiences at the University of Bologna, Mertek Engineering, Erna Mas, and NG Kütahya Ceramic, I successfully carried out an undergraduate research project funded by the Scientific and Technological Research Council of Türkiye (TÜBİTAK) on battery balancing systems for electric vehicles. I possess an IELTS Overall score of 8.0, international experience, strong communication skills, and a high level of work discipline.

### WORK EXPERIENCE

#### ERNA MAS

##### Engineer Intern

Istanbul, Turkey

August 2026

In the Technology and Software Development Department, I participated in the testing and validation processes of electronic boards, and was responsible for analyzing and reporting the test results. I contributed to hardware design and circuit development efforts, and prepared and delivered technical presentations for the developed designs. Furthermore, I worked on the design and development of buck-boost, flyback, and capacitive power supplies.

#### Mertek Engineering

##### Engineer Intern

Bursa, Turkey

October 2025 - January 2026

In the field of electrical contracting, I collaborated with Electrical Engineer Mert Kaya to create technical drawings for electrical installation and application projects using AutoCAD. During field operations, I inspected electrical installations to ensure their compliance with project specifications and standards, while managing the tracking and reporting of these processes. Additionally, I communicated directly with clients to analyze their technical needs, and played an active role in the quoting, proposal, and marketing processes.

#### Bologna University

##### Engineer Intern

Cesena, Italy

July 2025 - October 2025

I worked on electronic hardware development alongside Prof. Marco Tartagni's team. As part of this project, I contributed to developing an alternative, low-cost prototype for multispectral cameras. I designed a PCB to control and drive the LEDs emitting the specific target wavelengths analyzed in the project, ensuring high current stability. I took an active role in all hardware processes, ranging from component selection and circuit design to PCB layout and final prototype development.

#### NG Ceramic

##### Engineer Intern

Kutahya, Turkey

July 2024 - August 2024

I was involved in the inspection and control processes of the factory's electrical systems and production lines. As part of this role, I conducted damage and fault inspections on equipment, and observed fault response procedures alongside occupational safety practices. Additionally, I gained practical experience regarding the operating principles of these systems by analyzing the electrical project documents and circuit drawings of the production lines.

## PROJECTS

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### Active Battery Management System Design

As part of an undergraduate research project funded by the Scientific and Technological Research Council of Türkiye (TÜBİTAK) under the supervision of Dr. Aykut Bıçak, I conducted literature reviews and technical analyses on active battery balancing systems. I modeled and simulated various topologies in MATLAB/Simulink, and based on the results, I developed a novel Zero Current Switching (ZCS) quasi-resonant active battery balancing topology. I designed the PCB for this system using KiCad and managed the entire process, including component selection, procurement, PCB assembly, and prototype testing. I am currently preparing an academic paper based on the results obtained from this project.

### Multi Spectral Camera Design

As part of a project aimed at creating a cost-effective alternative to expensive multispectral cameras, I researched and selected suitable LED components capable of emitting the target wavelengths. I designed a TPL7407-based LED driver circuit to ensure the LEDs were driven at their targeted peak wavelengths with high current precision, and I validated the design through simulations in Proteus and TINA. Furthermore, to enable battery-powered operation, I selected appropriate DC-DC converters, designed the power management architecture, and completed the comprehensive PCB design for the entire system.

### 8 Bit Computer Simulation

To gain a deeper understanding of the fundamental working principles of computer systems, I developed an 8-bit computer architecture simulation in Proteus. Within this design, I simulated core processing and data storage structures by successfully implementing the storage of 8-bit data in RAM and the execution of operations through an Arithmetic Logic Unit (ALU).

## EDUCATION

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### Bursa Technical University

Electrical and Electronics Engineering, GPA: 3.09 / 4

Bursa, Turkey

October 2022 - July 2026

## TECHNICAL SKILLS

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### Circuit Design

Power Supply Design, EMI Filter Design, Driver Circuit Design, Circuit Analysis

### Technical Programs

PSpice / OrCAD, MATLAB / Simulink, KiCad, Proteus, LTspice, LaTeX

### Workshop Skills

Oscilloscope, Multimeter, Soldering

## LANGUAGES

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### Turkish

Mother tongue

### English

IELTS: C1 (Overall 8.0)

## REFERENCES

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### Levent GÖMÜRGEN

Erna MAS / Senior Hardware Engineer

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### Prof. Marco TARTAGNI

Bologna University / Professor

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### Dr. Aykut BIÇAK

Bursa Technical University / Associate Professor

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